Canonical notation, identifiers, and glossary

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This file is the single source of truth for mathematical notation, code identifiers, and controlled vocabulary across the manuscripts under `analysis/report/`. New manuscripts should adopt it rather than re-deriving symbols, and any deviation should be stated explicitly in the deviating paper's Methods section.

The audit behind it is `analysis/report/whitepaper-notation-audit.md`, whose PDF binds this file in as its appendix so that the two cannot drift apart.

Symbols and identifiers listed here were extracted from the manuscript sources, from `R/`, from the drivers under `analysis/scripts/`, and from the archived `.rds` summaries.

Part 1. Mathematical symbols

Indices and structure

Symbol	Meaning
i	participant index
t	timepoint index; weekly measurement occasions throughout
j, ij	period or occasion index within participant (papers 04, 06)
N	number of participants. **State whether per randomization path or in total**; both conventions are in use
n_{reps}, n_{sim}	Monte Carlo replicates per cell
k	cycles per participant in design sweeps; also the Weibull shape parameter in the decay family

Outcome and latent components

Symbol	Meaning
Y_{it}	observed symptom score; the modelled outcome (code column `Sx`)
BL_i	participant-specific baseline level
BR_{it}	biological (pharmacological) response component
PB_{it}	placebo-belief response component
TV_{it}	time-variant natural-history component
ε_{it}	observation-level residual
u_i	participant random intercept
η	expectancy covariate (measured belief), paper 06

Treatment exposure and carryover

Symbol	Meaning
D_{it}	binary drug state: 1 on drug, 0 off drug
$D_{bc,it}$	continuous exposure-decayed drug indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off drug (code column `Dbc`)
t_{sd}	time since discontinuation; input to the carryover decay (code `tsd`)

Symbol	Meaning
t_{od}	cumulative time on drug; input to the Gompertz response (code `tod`)
t_{pb}	time-by-placebo-belief accumulation (code `tpb`)
$t_{1/2}$	carryover half-life (code `carryover_t1half`)
λ	carryover decay rate, $\ln 2/t_{1/2}$ (code `lambda_cor`)
λ_w	Weibull rate, $(\ln 2)^{1/k}/t_{1/2}$

Decay families implemented on the data-generating side: exponential $e^{-\lambda t}$; linear $\max(0, 1 - t/(2t_{1/2}))$; Weibull $e^{-(\lambda_w t)^k}$; power $\max(0, (1 - t/(3t_{1/2}))^p)$.

Biomarker and moderation

Symbol	Meaning
B_i	participant-level pre-treatment biomarker value (code `bm`)
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$ (code `bm_z`)
c_{bm}	moderation parameter of the **covariance-moderation** architecture: on-drug correlation between biomarker and BR. A correlation, bounded in $[-1, 1]$
β_{bm}	moderation parameter of the **mean-moderation** architecture: dimensionless multiplier scaling the treatment effect by b_i
$c_{bm,A}$, $c_{bm,B}$	independent mean-channel and covariance-channel weights of the dual-channel architecture
σ_{BR}, σ_{bm}	standard deviations of the response component and of the biomarker

Estimand, inference, and performance

Symbol	Meaning
$\beta_{bm:D}$	the target estimand: the biomarker-by-treatment interaction coefficient (`bm:Dbc`)
β_{bm}^{BR}, β_{bm}^{PB}, β_{bm}^{TV}	component-specific biomarker slopes (paper 06)
$\beta_{bm}^{\text{lumped}}$	the one-component (single-indicator) biomarker slope
β_D	treatment main effect (papers 04, 05)
θ, $\hat{\theta}$	a generic interaction coefficient; used only where the argument is deliberately not N-of-1 specific (paper 10)
θ_{true}	calibrated true value, $-c_{bm}\,\sigma_{BR}$
ρ	AR(1) within-factor serial correlation
κ	ratio of mean model-based standard error to empirical standard deviation; $\kappa > 1$ means a conservative test
δ	standardized bias of the test statistic
ν	denominator degrees of freedom
α	nominal significance level; 0.05 throughout
π	power, $\Pr(p < \alpha)$
γ	latent-class gating slope (paper 03); a $BR \times PB$ interaction coefficient (paper 06)
Z_i	unobserved latent class label (paper 03)

The five rules the symbols encode

1. D_{it} **is binary;** $D_{bc,it}$ **is continuous.** These are different objects, and the distinction is the subject of the carryover-specification manuscript. Never use one for the other.
2. c_{bm} **is a correlation;** β_{bm} **is a multiplier.** They are calibrated to a common numeric scale, because the mean-channel shift is applied as $\beta_{bm}\, b_i\, \sigma_{BR}$ on the standardized biomarker, so the reference value 0.45 denotes a matched moderation strength under either architecture. The symbols are nonetheless not interchangeable. A paper reporting both architectures at a matched value may use one shared label, but must say so.
3. **Mathematics in model definitions, code identifiers in code contexts.** Write $Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \ldots$ for the model; write `Sx ~ bm + t + Dbc + bm:Dbc` in `\texttt{}` when quoting the R formula. Never mix the two inside one expression.
4. **State the sample-size convention.** N per randomization path and N in total are both in use and are not interchangeable; every Methods section must say which it means.
5. **State the sign convention.** The three components are non-negative *reductions* in symptom severity, so an increase in any component lowers Y. Treatment effects and interaction coefficients are consequently negative; moderation parameters are positive.

Units

Carryover half-life $t_{1/2}$ is quoted in **weeks** in the interaction-focused manuscripts, on the canonical grid $\{0, 0.5, 1.0\}$, and in **days** in the main-effect and test-procedure manuscripts, on a pharmacokinetic scale with a 3-day baseline. Both conventions are retained because each paper's figures are drawn on its own scale. Every paper using days carries an explicit units paragraph giving the conversion (1 week = 7 days). Timepoints t are weekly in both conventions.

Part 2. Code identifiers and stored data values

These are the names as they appear in `R/`, in the drivers, and in the archived `.rds` summaries.

Where a stored value is also a reporting label, the two are kept **identical**, so that no display-time mapping stands between the data and the page. A mapping layer is a standing hazard: it has to be applied at every display site, it fails silently when a rename reaches one layer and not the other, and it makes a stored table unreadable without the manuscript beside it. Where a stored value is a package API argument (`dgp_architecture = 'mvn'`), it stays as the API defines it and the manuscript maps it to a reporting name, which is the ordinary factor-level-to-label relationship.

Exported package functions

`buildtrialdesign`, `buildSigma`, `generateData`, `censordata`, `lme_analysis`, `generateSimulatedResults`, `validateParameterGrid`, `cumulative`, `modgompertz`, `trajectoryShape`, `shape_logistic`, `shape_hyperbolic_tangent`, `shape_piecewise_linear_breakpoint`, `characterize_carryover`, `analyze_trial_extended`, `print_carryover_summary`, `print_trial_summary`, `plotfactortrajectories`, `PlotModelingResults`, `reknitsimresults`.

Principal arguments

Identifier	Role
`modelparam`	list: sample size `N`, moderation `c.bm`
`respparam`	Gompertz parameters `maxr`, `rate`, `disp` per component
`blparam`	biomarker and nuisance means and standard deviations
`trialdesign`	output of `buildtrialdesign`: `timepoints`, `timeptnames`, `expectancies`, `ondrug`
`dgp_architecture`	`'mvn'`, `'mean_moderation'`, or `'combined'`
`carryover_t1half`	carryover half-life

Identifier	Role
`lambda_cor`	decay rate; derived from the half-life when `NA`
`br_family`	response-curve family: `'gompertz'` and alternatives
`br_p2`, `br_p3`	shape parameters of the alternative families
`moderation_scaling`	`'constant'` or trajectory-scaled
`cached_sigma`	pre-built covariance matrix, reused across parallel workers
`makePositiveDefinite`	positive-definiteness repair switch
`n_cores`	worker count; `-1` auto-detects
`save_chunks`	progressive checkpointing for long runs

Data columns

`ptID` (participant), `Sx` (outcome, the Y_{it} above), `bm` (biomarker), `bm_z` (standardized biomarker), `Dbc` (continuous exposure indicator, $D_{bc,it}$), `tsd`, `tod`, `tpb`, `t_wk` (time in weeks), `path` (randomization path).

Stored factor levels and their reporting labels

Column	Stored values	Reported as	Mapped?
`spec`	`E1`, `E2`, `E3`	E1, E2, E3	no, identical
`carryover_form`	`exponential`, `linear`, `weibull`	as stored	no, identical
`design`	`CO`, `Hybrid`, `OLBDC`	CO, Hybrid, OL+BDC	only `OLBDC` to `OL+BDC`
`dgp_arch`	`mean_moderation`, `mvn`, `combined`	Mean, Covariance, Dual	yes; these are package API values

The `spec` column stored `A1`, `A2`, `A3` until 2026-07-31. Twenty-one `.rds` files were migrated to `E1/E2/E3` and the producing drivers updated to emit them, so the display-time mapping that previously bridged the two has been retired. Pre-migration copies are in `analysis/.spec-migration-backup/`. Output archived before that date, including anything under a paper's own `archive/`, still uses the old values.

Summary-table columns

`n_reps`, `n_converged`, `converged_frac`, `non_convergence_rate`, `power`, `bias`, `mean_estimate`, `true_value`, `empirical_se`, `mse`, `coverage`, and an `mc_se_`-prefixed Monte Carlo standard error for each (`mc_se_power`, `mc_se_bias`, `mc_se_mse`, `mc_se_coverage`, `mc_se_non_convergence`).

Part 3. Controlled vocabulary and glossary

The project lexicon at `docs/29-nof1-precision-medicine-lexicon.md` holds the full terminology set (241 terms across 13 thematic sections, with cross-field synonyms). The entries below are the subset carrying a fixed, enforced meaning in this compendium.

Trial designs

OL (open-label). Unblinded titration; no off-drug contrast, so it supplies no within-subject interaction information. **BDC** (blinded discontinuation). A blinded switch to placebo after open-label response. **OL+BDC**. Open-label titration followed by blinded discontinuation. **CO** (crossover). Traditional two-period within-subject crossover. **Hybrid**. The Hendrickson (2020) design: open-label titration, blinded discontinuation, then a brief crossover. The programme's reference design. Capitalize it. **Aggregated N-of-1 trial**. Many single-patient on/off series pooled into one mixed model.

Define each abbreviation at first use in every paper.

Data-generating architectures

Named by the channel that carries the interaction. Single letters are no longer used.

Name in prose	Legends and tables	Data value	Former label
the mean-moderation architecture	Mean	`mean_moderation`	Architecture A
the covariance-moderation architecture	Covariance	`mvn`	Architecture B
the dual-channel architecture	Dual	`combined`	Architecture C

Mean-moderation. The biomarker scales the treatment effect in the conditional mean; the interaction lives in the first moment. The near-universal convention in the trial-simulation literature. **Covariance-moderation.** The biomarker-response correlation is treatment-state dependent; the interaction lives in the second moment. The Hendrickson formulation, and the one whose power is most eroded by carryover. **Dual-channel.** Both channels active with independent weights $c_{bm,A}$ and $c_{bm,B}$; the other two are its single-channel limits. A data-generating construct only: the two weights are not separately identifiable from a fitted `bm:Dbc` coefficient.

Analysis specifications for carryover

E1. Binary on-drug indicator; carryover ignored. **E2.** Exposure-weighted continuous predictor D_{bc}, committing to an assumed decay form and half-life. **E3.** Binary indicator plus a lagged just-off-drug nuisance term; the classical Jones-Kenward crossover device, assuming no decay form.

The letter E stands for the exposure regressor that distinguishes them. These were labeled A1/A2/A3 until 2026-07-31; the stored data and the drivers were migrated to E1/E2/E3, so the labels now agree everywhere and nothing is mapped at display time.

Why not A/B/C and M1/M2/M3. The original scheme used A/B/C for architectures and A1/A2/A3 for specifications, colliding on the letter A within a single manuscript. Relabeling the specifications M1/M2/M3 removed that collision but created a second one: the calibration manuscript already uses M0-M3 for its working-covariance model ladder, and those labels are keys into archived data. E1/E2/E3 is free across the whole corpus, and naming the architectures by mechanism removes the class of collision entirely.

Response-curve families

Modified Gompertz (default), $y = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t))$ with a vertical offset so the curve passes through the origin. Alternatives: **logistic** (symmetric sigmoidal), **hyperbolic tangent** (fast-saturating), **piecewise-linear breakpoint** (no smoothness assumption).

Missing data and dropout

MCAR, **MAR**, **MNAR** in the standard Rubin sense. **Informative dropout.** Hazard depending on cumulative symptom worsening since baseline. **Happy accident.** The randomization-path selection effect by which informative dropout preferentially preserves the most informative crossover blocks.

Inference

LME. Linear mixed-effects model; the programme's default analysis. **corCAR1.** Continuous-time AR(1) residual correlation structure. **GEE.** Generalized estimating equations; the marginal comparator. **Mancl-DeRouen**, **CR2.** Small-sample bias-corrected sandwich variance estimators. **RM-ANOVA.** Repeated-measures analysis of variance; the strict form requires a dichotomized biomarker. **Anti-conservative.** Realized Type I error above nominal. **Working covariance.** The assumed residual covariance of the LME; reserve **working correlation structure** for the GEE object.

Estimands and reporting

Biomarker-treatment interaction. The programme's core estimand; equivalently the patient-by-treatment interaction, or heterogeneity of treatment effects (HTE) in the precision-medicine literature. **Predictive biomarker.** Predicts differential benefit, as opposed to a **prognostic** biomarker, which predicts outcome regardless of treatment. **ADEMP.** The Morris, White and Crowther (2019) reporting framework: aims, data-generating mechanisms, estimands, methods, performance measures. **MCSE.** Monte Carlo standard error; reported with every performance measure.

Enumeration conventions

Simulation studies. Number them (Study 1, Study 2, …) rather than lettering them. **Sensitivity sweeps.** Labels S1, S2, … are paper-local and are not comparable across papers. Cite them with the paper prefix when crossing manuscripts (05-S2, not S2).

Prose terminology

Symbol conventions are settled here; prose terminology is settled by `docs/30-terminology-consistency-plan.md`, executed 2026-07-30 with its D1 spelling decision locked to **US English**, overriding that document's original frequency-based recommendation of British forms. The two are complementary: this file governs mathematics and identifiers, that one governs surface forms and acronym expansion.

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